Music Preference	Introverts	Extroverts	TOTAL
POP	9	9	18
INDIE/ALTERN.	8	4	12
RAP/HIP-HOP	2	3	5
ROCK	3	3	6
R&B	0	2	2
COUNTRY	0	1	1
CLASSICAL	0	1	1
REGGAETON	1	6	7
	23	9	52

variable 1

variable 2

Music Preference	Introverts	Extroverts	TOTAL
POP	9 (7.9615)	9 (10.0385)	18
INDIE/ALTERN.	8 (5.3077)	4 (6.6923)	12
OTHER	6 (9.7308)	16 (12.2692)	22
TOTAL	23	29	52

1. Communicate

Association b/t music preference & personality type

INFERENCE: Chi-square test of independence

H_0: Music preference and personality type ARE independent

H_a: Music preference and personality type ARE associated

2. Conditions

1) Random: multistage random sample ✓
2) Independent: $n \leq 10\%$ (Doral Academy students); $n=52$ ✓
3) Large counts: $E = \dfrac{(\text{row total})(\text{column total})}{\text{grand total}} \geq 5$ EACH ✓

As all conditions are satisfied, we may proceed.

3. Calculations

$$x^2 = \sum \frac{(O-E)^2}{E}$$

$$x^2 = \frac{(9-7.9615)^2}{7.9615} + \frac{(9-10.0385)^2}{10.0385} + \frac{(8-5.3077)^2}{5.3077} + \frac{(4-6.6923)^2}{6.6923} +$$

$$\frac{(6-9.7308)^2}{9.7308} + \frac{(16-12.2692)^2}{12.2692} = 5.2572 \rightarrow \underline{\text{test statistic}}$$

$df = (r-1)(c-1)$

$(3-1)(2-1) = (2)(1) = \underline{2} - df$

$x^2cdf(5.2572, \infty, 2) = 0.0721$

$$\boxed{\text{p-value} = 0.0721}$$

4. Conclusion

As our p-value of $0.0721 > 0.05$ (α), we FAIL TO REJECT the null hypothesis. Therefore, we do not have sufficient statistical evidence at the 5% α that music preference and personality type are associated.

2-Sample T-Test for Difference of Means

1. Communicate

Groups:

Introverts : extroversion score < 50

Extroverts : extroversion score > 50

parameters:

μ_I = true mean # of hours per day listening to music for introverts

μ_E = true mean # of hours per day listening to music for extroverts

$H_0: \mu_I - \mu_E = 0$

$H_a: \mu_I - \mu_E \neq 0$

Inference: 2-Sample T-Test for Difference of Means

2. Conditions

1.) Random: multistage random sampling

2.) Independence: $n \leq 10\%$ (Doral Academy students); $n = 52$

3.) Normal: $n = 52$; $52 > 30$

As all conditions are satisfied, we may proceed.

3. Calculations

Introverts $(x < 50)$ | $n_I = 23$ | $\frac{\overset{(84)}{\sum \text{hours}}}{23} = \bar{x}_I = 3.5$

Extroverts $(x > 50)$ | $n_E = 29$ | $\frac{\overset{(102)}{\sum \text{hours}}}{29} = \bar{x}_E = 3.5172$

Sample s:

$s_I \approx 2.770$

$s_E \approx 2.126$

Summary stats:

	n	$\bar{x}$	s
Intro :	24	3.5	2.770
Extro :	29	3.517	2.126

df ≈ 42

$$t = \frac{(\bar{x}_I - \bar{x}_E) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_E^2}{n_E}}} = \frac{(3.500 - 3.51724)) - 0}{\sqrt{\frac{(2.770)^2}{24} + \frac{(2.126)^2}{29}}} \approx -0.0250$$

$p(|t| \geq 0.025) \approx 0.980$

4. Conclusion

As our p-value of 0.980 is much greater than $\alpha = 0.05$, we fail to reject the null hypothesis. Therefore, there is not sufficient statistical evidence to conclude that introverts and extroverts differ in their mean number of hours listening to music per day in the population of students like those sampled.